Anisotropic Hubble expansion from an off-center observer in a Sedov–Taylor blast-wave universe

C. A.

2026-09-19

Abstract

We synthesize a Newtonian working model in which the Universe expands as a Sedov–Taylor (ST) homologous blast about a *real* geometric center of expansion (CoE) beyond the observable horizon, with the observer off-center. Homology alone yields an isotropic apparent Hubble parameter; anisotropy requires a homology-breaking mechanism. We take that mechanism to be Gauss / shell-theorem gravity from a centrally concentrated, radially decreasing density $\rho(r)$, so $g = -GM(< r)/r^2$ (not a uniform ball). Preferring time-integrated peculiar velocities over a bolt-on impulse $u \sim \alpha g/H$, ideal integration only rescales the amplitude ($\alpha \to \alpha_{\rm eff}$) and does *not* flip the angular shape of the Migkas-style normalization $A(\varphi)$; expand versus collapse does. Mapping $A_{\rm hratio} \propto H_{\rm obs}/H$ against the Migkas et al. (2020) A versus Galactic longitude pattern, first-principles ST+Gauss geometry alone reaches $R^2 \approx 0.543$. A diagnostic Amp$\times$ sin phase rotation raises $R^2 \approx 0.746$; a multiplicative 2ℓ residual layer reaches $R^2 \approx 0.875$ (RMSE 0.038 $\to$ 0.029). Amp$\times$ sin and the 2ℓ multiplier are phenomenological; the first-principles claim is the geometry. The CMB remains the principal adversary (spherical off-center voids are dipole-limited to ~ 15 Mpc). This paper does not claim to have falsified the cosmological principle.

Contents

1 Introduction **3**

2 Model: Sedov–Taylor background, Gauss gravity, off-center observer **3**
2.1 Geometry . . . 3
2.2 Homologous Sedov–Taylor background . . . 4
2.3 Gauss gravity with decreasing density . . . 4

3 Bolt-on peculiar velocity versus ideal time-integrated gravity **5**
3.1 Bolt-on impulse . . . 5
3.2 Ideal time integration . . . 5

4 From $H_{\rm obs}$ to $A(\varphi)$ and Migkas Fig. 6 **5**

5 Fits: level sets, R^2 ladder, and residuals **6**
5.1 Geometry grid (first principles) . . . 6
5.2 R^2 ladder (honesty about knobs) . . . 6

6 Physical reading of Amp$\times$ sin, 2ℓ, and nesting dolls **7**

7 Related work **9**

8 Discussion: what a good fit does and does not prove **10**
8.1 What $R^2 \approx 0.54$–0.88 means 10
8.2 CMB adversary 10
8.3 Falsifiers 11
8.4 Open problems 12

9 Conclusions **12**

1 Introduction

The cosmological principle (CP) asserts statistical homogeneity and isotropy on large scales: no preferred center, and a suitably defined Hubble expansion that is isotropic about every comoving observer. Late-time directional probes contest that claim at the several-to-tens-of-percent level. In particular, Migkas et al. (Migkas et al., 2020) reported that the normalization A of the X-ray cluster L_X–T relation varies with Galactic longitude (their Fig. 6), with a preferred direction near $(\ell, b) \sim (280^\circ, -20^\circ)$ and a maximum luminosity-distance variation of order $\sim 16 \pm 3\%$ under an isotropic-scaling interpretation. Those results remain contested—bulk flows, absorption, calibration, and scatter can all mimic directional signals—but they supply the empirical sky pattern against which any centered alternative must be aimed.

The program synthesized here is stronger than a mild ΛCDM tweak. It asserts:

1. The Universe has a **real center of expansion (CoE)**.
2. That CoE lies **beyond the observable horizon**.
3. We are an **off-center observer** at Euclidean (Newtonian) distance D from the CoE.
4. Expansion is usefully modeled, at the Newtonian skeleton level, by a **Sedov–Taylor blast** with a **radially decreasing** density $\rho(r)$ about the CoE; gravity follows Gauss's law with enclosed mass $M(< r)$ identified with the mass between us and the CoE.

The scientific goal is to locate that center—or kill the idea if CMB isotropy and related probes make a real center untenable (Alnes and Amarzguioui, 2006; Kodama et al., 2010; Saadeh et al., 2016; Planck Collaboration, 2020).

Historical cousins include Fennelly's off-center Hubble anisotropy (Fennelly, 1977), Bertschinger's cosmological ST shocks (Bertschinger, 1983), Smoller–Temple shock cosmologies inside / beyond a Hubble-scale black hole (Smoller and Temple, 1998, 2003), and the LTB off-center CMB literature (Alnes and Amarzguioui, 2006; Kodama et al., 2010; Blomqvist and Mörtsell, 2010). None combines ST+Gauss geometry, a beyond-horizon CoE, and a Migkas $A(\ell)$ target with explicit CMB accounting; that gap is the present niche.

2 Model: Sedov–Taylor background, Gauss gravity, off-center observer

2.1 Geometry

Place the CoE at the origin. The observer sits at $\vec{D}$ with $D = \|\vec{D}\|$. A source is at $\vec{r}$ (CoE-radius $r = \|\vec{r}\|$), with observer–source separation $\vec{d} = \vec{r} - \vec{D}$, $d = \|\vec{d}\|$, and line of sight $\hat{n} = \vec{d}/d$. The preferred axis $\hat{D} = \vec{D}/D$ points from CoE to observer; the direction from observer toward the CoE is $-\hat{D}$.

Toy calculations place sources on an observer-centered circle of fixed d with Earth-centric angle φ measured from $+\hat{D}$ (anti-center):

$$\hat{n} = (\cos\varphi, \sin\varphi), \qquad \mu \equiv \hat{n} \cdot \hat{D} = \cos\varphi. \tag{1}$$

Thus $\varphi = 0$ looks away from the CoE and $\varphi = \pi$ looks toward it. When comparing to Galactic longitude ℓ, an overall sky-axis orientation is absorbed into a phase (Sec. 5). Scale lengths entering the mass profile are denoted L (or r_s); the dimensionless ratios of interest are d/D, D/L, and M/M_0 (mass in units of a reference M_0).

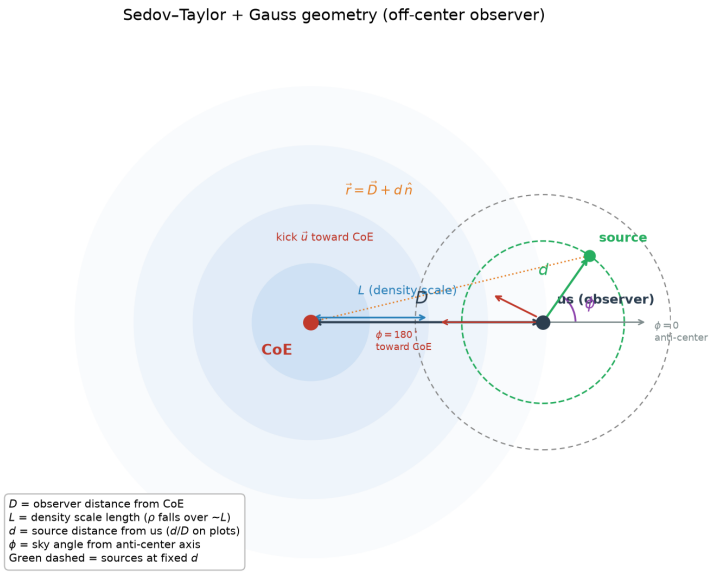


Figure 1: Geometry of an off-center observer at distance D from a real CoE, with source distance d and angle φ from the preferred axis. First-principles ST+Gauss kinematics are computed in this plane (then rotated on the sky as a diagnostic).

2.2 Homologous Sedov–Taylor background

Classical Sedov–Taylor blast front in ambient density ρ_0 with energy E:

$$R_{\rm ST}(t) = \xi\Big(\frac{E}{\rho_0}\Big)^{1/5} t^{2/5}, \qquad H(t) \equiv \frac{\dot{R}_{\rm ST}}{R_{\rm ST}} = \frac{2}{5t}. \tag{2}$$

Interior working assumption (homologous expansion or collapse):

$$\vec{v}(\vec{r}, t) = H_{\rm bg}\,\vec{r}, \qquad H_{\rm bg} = +H \quad \text{(expand)} \quad \text{or} \quad -H \quad \text{(collapse)}. \tag{3}$$

Observer and source velocities in the CoE frame are then $\vec{v}_O = H_{\rm bg}\vec{D}$ and $\vec{v}_S = H_{\rm bg}\vec{r}$, so

$$\vec{v}_{\rm rel} = H_{\rm bg}\,\vec{d}, \qquad v_{\rm los} = \vec{v}_{\rm rel}\cdot\hat{n} = H_{\rm bg}\,d, \qquad H_{\rm obs} = \frac{v_{\rm los}}{d} = H_{\rm bg}. \tag{4}$$

Critical kinematic result: pure homology yields $H_{\rm obs} = H_{\rm bg}$ *independent of direction*, for any offset D. Geometry alone does not produce Migkas-like anisotropy. Anisotropy requires breaking homology (gravity, non-homologous ST structure, or relativistic / light-cone effects). Cosmological identification of (E, ρ_0, ξ) remains undefined; Eq. (2) is a similarity scaffold, not a derivation of FLRW (Bertschinger, 1983).

2.3 Gauss gravity with decreasing density

Under spherical symmetry, Newton's shell theorem / Gauss's law gives

$$\vec{g}(\vec{r}) = -\frac{G\,M(<r)}{r^2}\,\hat{r}, \qquad M(<r) = \int_0^r 4\pi s^2 \rho(s)\,ds. \tag{5}$$

The working density is **centrally concentrated and radially decreasing**. A convenient Phase A profile (incomplete-gamma cumulative of $\rho \propto e^{-r/L}$) is

$$\frac{M(<r)}{M} = 1 - e^{-X}\Big(1 + X + \frac{X^2}{2}\Big), \qquad X = \frac{r}{L}. \tag{6}$$

Rejected special case: uniform ball $\rho = \text{const} \Rightarrow M(< r) \propto r^3 \Rightarrow g \propto -r$, which produces only a monopole shift in $H_{\rm obs}$ and no sky anisotropy. Decreasing ρ makes $M(< r)$ grow slower than r^3, retaining tidal $1/r^2$ structure near $r \sim D$.

Chris identifies the relevant enclosure with the mass toward the CoE. At observer scale the dimensionless strength is

$$\kappa \equiv \frac{G\,M(< D)}{D^3 H^2}. \tag{7}$$

For an $\Omega_m \sim 0.3$ critical sphere one has the natural scale $\kappa_{\rm natural} \sim \Omega_m/2 \sim 0.15$; the program's working $\kappa \sim \mathcal{O}(0.1\text{–}1)$ relative to $M_{\rm crit}(< D)$ is meant to be natural, not absurd.

3 Bolt-on peculiar velocity versus ideal time-integrated gravity

3.1 Bolt-on impulse

Acceleration is not velocity. A common order-of-magnitude ansatz is an impulse over a Hubble time,

$$\vec{u}(\vec{r}) = -\alpha\,\frac{G\,M(< r)}{r^2 H}\,\hat{r}, \qquad \alpha \sim \mathcal{O}(1), \tag{8}$$

with relative peculiar contribution

$$\delta v = \left[\vec{u}(\vec{r}) - \vec{u}(\vec{D})\right] \cdot \hat{n}, \qquad H_{\rm obs} = H_{\rm bg} + \frac{\delta v}{d}. \tag{9}$$

On an observer-centered circle of radius $d = \varepsilon D$ (code units often take $\varepsilon \sim$ few $\times\, 10^{-1}$), Eq. (9) is the closed form used throughout the project's forward maps (`H_obs_circle`).

3.2 Ideal time integration

Preferring physics from the blast start, one integrates g along homologous trajectories. Three Newtonian treatments yield effective bolt-on amplitudes $\alpha_{\rm eff}$ (Bertschinger, 1983, cf. project notes):

1. **Frozen path:** $\Delta v = \int_{t_i}^{t_f} g(r_{\rm bg}(t))\,dt \Rightarrow \alpha_{\rm eff} = 2\big(1 - \eta^{1/5}\big)$ with $\eta = t_i/t_f$ (approaches 2 as $\eta \to 0$).
2. **Hubble drag:** $\mathrm{d}u/\mathrm{d}t + Hu = g \Rightarrow \alpha_{\rm eff} = (2/3)\big(1 - \eta^{3/5}\big)$ (approaches 2/3).
3. **Nonlinear ODE:** $\ddot{r} = -(6/25)r/t^2 - GM/r^2$ (ST background acceleration plus gravity).

Key structural result: time integration only rescales amplitude ($\alpha \to \alpha_{\rm eff}$). It does *not* invent a new small-κ regime, and it does *not* flip the angular form of $A(\varphi)$. Only the sign of the background expansion—expand ($+H$) versus collapse ($-H$)—flips the trough/peak morphology after the nonlinear A map (Fig. 2).

4 From $H_{\rm obs}$ to $A(\varphi)$ and Migkas Fig. 6

Migkas et al. (Migkas et al., 2020) map the L_X–T normalization A in $\Delta\ell = 90°$ Galactic-longitude strips. Under isotropic-scaling assumptions, A tracks luminosity distance (hence inverse powers of an effective H). Two mean-renormalized proxies are used in this program:

$$A_{\rm hratio} \propto \frac{H_{\rm obs}}{H} \qquad \text{(primary; Chris sheet “AF”)}, \tag{10}$$

$$A_{\rm invsq} \propto \frac{1}{(H_{\rm obs}/H)^2} \qquad (L_X\text{–}T \;/\; D_L\text{-motivated alternative}). \tag{11}$$

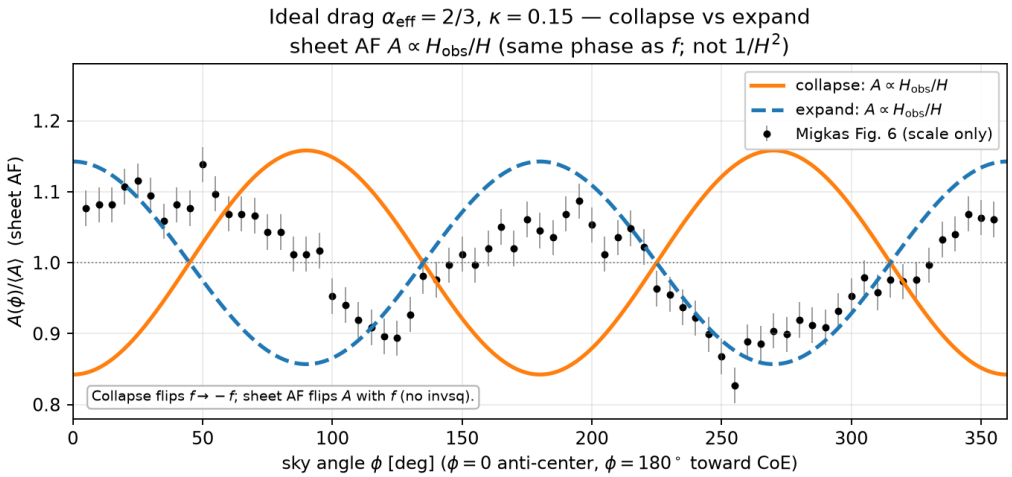


Figure 2: Collapse versus expand under ideal Hubble drag, mapped with sheet AF $A \propto H_{\rm obs}/H$ (mean-renormalized; Migkas points for scale only). Opposite $H_{\rm bg}$ flips the trough/peak morphology; there is no $1/H^2$ inversion here. Ideal integration only rescales amplitude. No Amp$\times$ sin / 2ℓ layers.

Both are then optionally smoothed with a wrap-around $\Delta\ell = 90°$ boxcar to match Migkas's strip geometry. Empirically, $A_{\rm invsq}$ *inverts* troughs and peaks relative to $A_{\rm hratio}$; sheet comparisons therefore prefer Eq. (10).

Closed forms for the circle map follow Eqs. (8)–(9) (or the same spatial structure with $\alpha \to \alpha_{\rm eff}$). Residual maps $\Psi = H_{\rm obs} - H$ vanish identically when $M = 0$ or $\alpha = 0$ (homology regression test).

5 Fits: level sets, R^2 ladder, and residuals

Correlation / affine R^2 is computed against digitised Migkas $A(\ell)$ along the same chain that produced `best_fit_2ell_multiplier.npz`.

5.1 Geometry grid (first principles)

Scanning $(M/M_0,\, d/D,\, L)$ on ideal ST+Gauss+$A_{\rm hratio}$ maps, the top geometry without Amp$\times$ sin rotation sits near

$$\frac{M}{M_0} = 3, \qquad \frac{d}{D} = 0.5, \qquad L \approx 0.169 \quad \big(D/L \approx 5.9\big), \tag{12}$$

with associated $\kappa \sim \mathcal{O}(0.7)$. Ideal ST+Gauss geometry alone yields

$$R^2 \approx 0.543 \tag{13}$$

against Migkas $A(\ell)$—a nontrivial longitude-pattern match, but far from a closed case.

5.2 R^2 ladder (honesty about knobs)

Amp$\times$ sin rotation (diagnostic). Aligning the model to Galactic longitude with a sheet-style phase factor,

$$A_{\rm rot}(\ell) = A(\ell) + \text{Amp}\,\sin(\omega\ell + \phi), \tag{14}$$

with best Amp / ϕ level

$$\text{Amp} = -0.06, \qquad \phi \approx 0.20\,\text{rad} \quad (R^2 \approx 0.746). \tag{15}$$

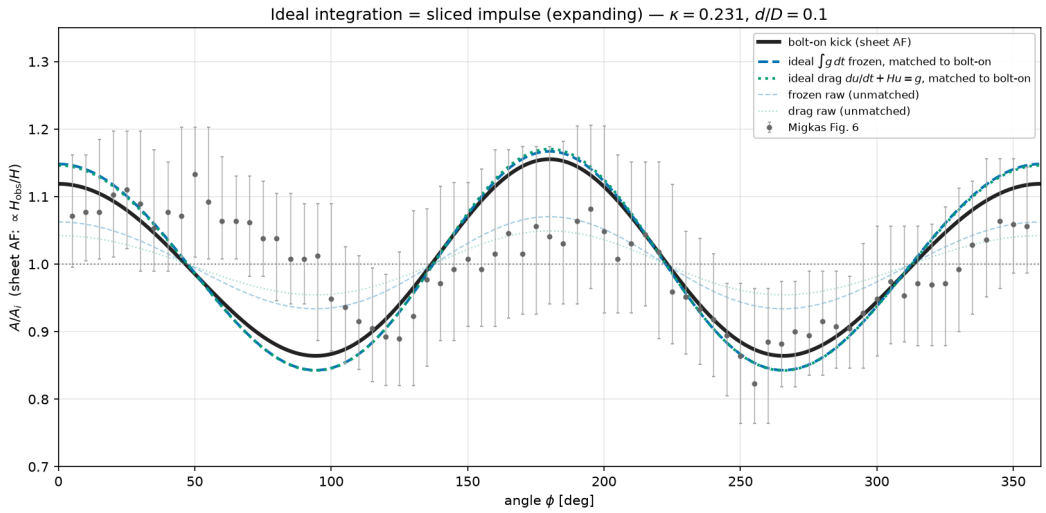


Figure 3: Ideal time-integrated ST+Gauss $A(\varphi)$ in the sheet AF ($A_{\rm hratio} \propto H_{\rm obs}/H$) convention, compared to digitised Migkas Fig. 6 longitude bars. Illustrative expanding slice ($\kappa \approx 0.231$, $d/D = 0.1$); bolt-on versus matched ideal drag / frozen curves coincide in angular shape. First-principles geometry only—no Amp$\times$ sin or 2ℓ multiplier.

Table 1: R^2 ladder versus digitised Migkas $A(\ell)$. First row is first-principles; subsequent rows add diagnostic / phenomenological layers.

Layer	Free knobs	R^2	RMSE
Ideal ST+Gauss geometry alone	geometry $(M, d/D, L)$	≈ 0.543	—
+ Amp$\times$ sin phase rotation	Amp, ϕ	≈ 0.746	0.038
+ multiplicative 2ℓ	ε, ψ	≈ 0.875	0.029

(The same chain's residual-2ℓ file used $\phi_1 \approx 0.30$ with Amp$= -0.06$ as a nearby local maximum; both sit on the Amp$= -0.06$ ridge.)

Multiplicative 2ℓ residual (phenomenological). Leftover quadrupole structure after Amp$\times$ sin is absorbed by

$$A \;\leftarrow\; \big(1 + \varepsilon \sin(2\ell + \psi)\big)\, A, \tag{16}$$

with best values from the project's NPZ chain

$$\varepsilon \approx -0.043, \qquad \psi \approx 176^\circ, \tag{17}$$

raising

$$R^2 :\; 0.746 \to 0.875, \qquad \text{RMSE} :\; 0.038 \to 0.029. \tag{18}$$

Honesty clause: Amp$\times$ sin and the 2ℓ multiplier are *not* derived from ST+Gauss alone. They are diagnostic layers (phase from possible non-radial observer velocity or axis misalignment; 2ℓ as leftover quadrupole). The first-principles claim is the geometry that already achieves $R^2 \approx 0.54$ before knobs. Reporting $R^2 \approx 0.88$ without that decomposition would be cheerleading.

6 Physical reading of Amp$\times$ sin, 2ℓ, and nesting dolls

Amp$\times$ sin. A pure radial peculiar field $\vec{u} \propto -\hat{r}$ about the CoE produces a fixed angular template once $\hat{D}$ is chosen. An additional non-radial observer velocity (or a misidentified preferred

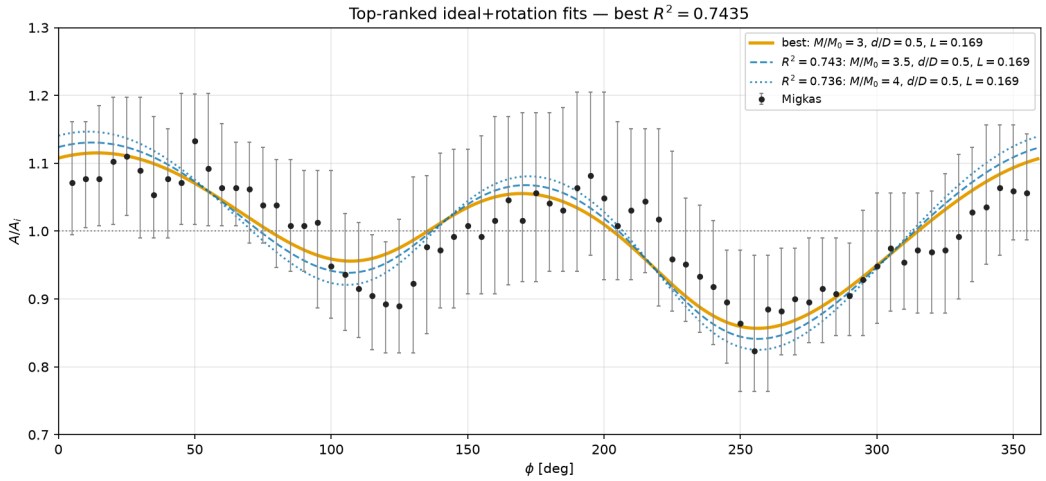


Figure 4: Top geometry-grid curves from the ideal-integration $(M/M_0, d/D, L)$ scan after Amp$\times$ sin alignment (diagnostic). Best raw geometry alone is $R^2 \approx 0.543$ ($M/M_0 = 3$, $d/D = 0.5$, $L \approx 0.169$); with rotation the grid peak is $R^2 \approx 0.74$.

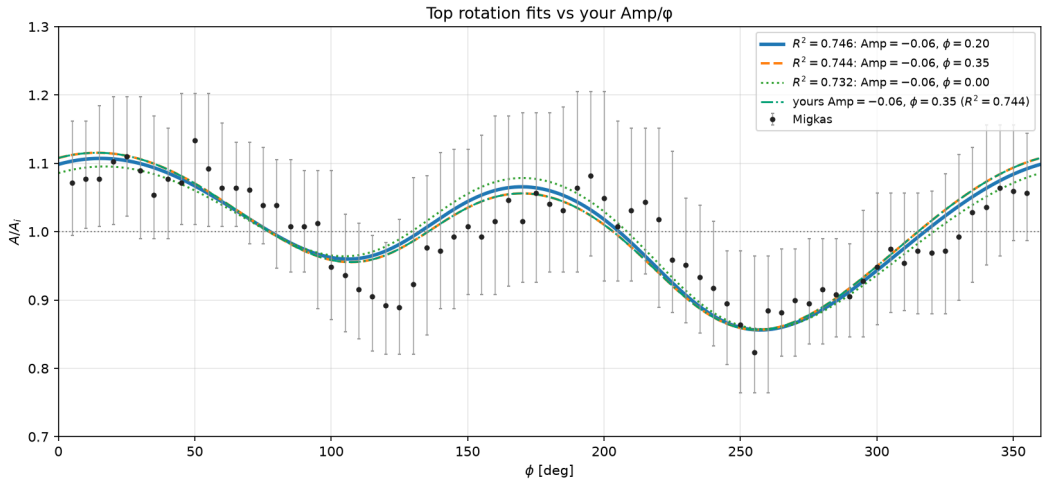


Figure 5: Amp$\times$ sin scan: top curves in (Amp, ϕ). Diagnostic layer only—phase can encode non-radial observer velocity or sky-axis misalignment, not a first-principles prediction.

axis on the sky) mixes a sinusoidal longitude modulation into the observed $A(\ell)$. The Amp$\times$ sin layer is the minimal phenomenological proxy for that mixing. It should not be read as evidence that ST+Gauss "predicts" Amp$= -0.06$.

2ℓ multiplier. At $d \ll D$ (CoE beyond the horizon, clusters at hundreds of Mpc), the linear tidal map of a spherical Gauss field is dominated by a *quadrupole* $P_2(\mu)$, not a pure dipole (Heinesen, 2021). Migkas Fig. 6 is a one-dimensional longitude projection; footprint leakage can project quadrupole power into an apparent dipole-like strip pattern. The multiplicative 2ℓ term absorbs residual quadrupole after the leading geometric template and Amp$\times$ sin are fixed. Again: diagnostic, not first principles.

Nesting dolls (speculation). Optional outer off-center masses—"nesting dolls" of additional centers—could source leftover multipoles beyond 2ℓ. This is marked as **speculation**: it is not required by the R^2 ladder, is not constrained by the present Newtonian pass, and must not be

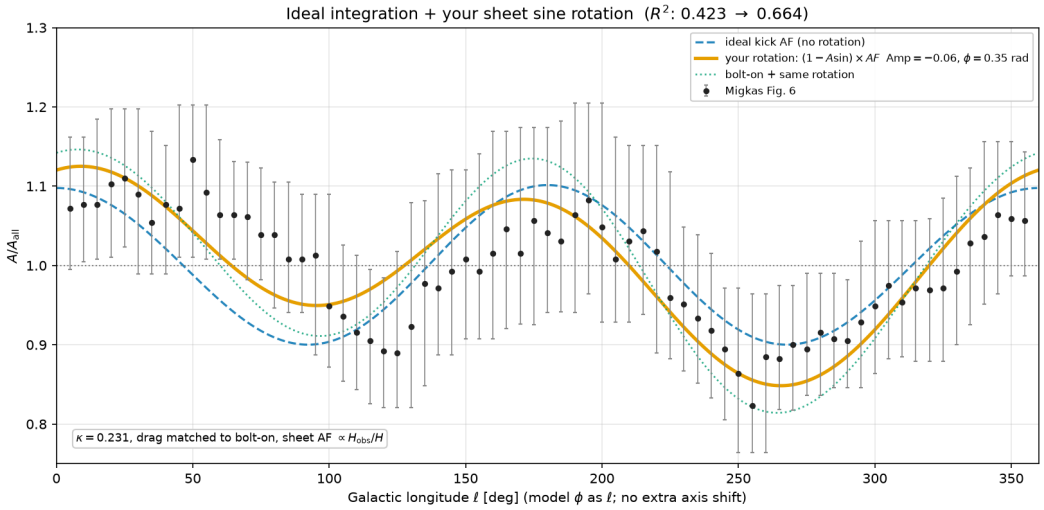


Figure 6: Sheet-style ideal ST+Gauss $A_{\rm hratio}$ (AF) after Amp× sin rotation (Amp= −0.06, $\phi \approx 0.35$ rad on this illustrative κ=0.231, d/D=0.1 slice) overlaid on Migkas bars. Geometry is first-principles; the rotation is diagnostic. This panel is *not* the Table 1 peak ($R^2 \approx 0.746$ at $\phi \approx 0.20$ rad on the best geometry).

sold as part of the ST+Gauss core.

7 Related work

Fennelly (Fennelly, 1977) fitted reported Hubble anisotropy to a tilted self-similar cosmology with a Gaussian density inhomogeneity, placing us ~ 120 Mpc off-center. Conceptual cousin (off-center observer, CMB-compatibility concern); not ST+Gauss, not Migkas $A(\ell)$.

Debney (Debney, 1977) / Pathria (Pathria, 1972) / Khakshournia (Khakshournia, 2014). "Universe as a black hole" / beyond-horizon matching language. Khakshournia shows Pathria-style null matching is not smooth (surface pressure on a null shell)—a warning against naive horizon identification.

Bertschinger (Bertschinger, 1983). Closest classical ST-in-cosmology dynamics: Sedov-like blast morphing into a thin dense shell in EdS. Supplies blast DNA, not off-center Migkas matching.

Smoller & Temple (Smoller and Temple, 1998, 2003). Exact FRW–TOV shock matching, including a Big Bang as a localized explosion inside an asymptotically flat Schwarzschild black hole with shock beyond a Hubble length. Closest GR cousin for beyond-horizon CoE ontology; continuum shock matching rather than time-integrated Gauss g.

Alnes & Amarzguioui (Alnes and Amarzguioui, 2006); Kodama et al. (Kodama et al., 2010); Blomqvist & Mörtsell (Blomqvist and Mörtsell, 2010). Off-center LTB voids: CMB dipole forces the observer within $\sim$ 15–16 Mpc of the center; SNIa alone are much weaker. Transfers the CMB-adversary toolkit to any spherical CoE.

Heinesen (Heinesen, 2021). Model-independent multipole decomposition of the anisotropic d_L Hubble law (9 / 25 / 61 dof at $\mathcal{O}(z)$ / $\mathcal{O}(z^2)$ / $\mathcal{O}(z^3)$). Language layer for forecasting ST+Gauss $d_L(\hat{n}, z)$ without committing to FLRW.

Saadeh et al. (Saadeh et al., 2016). Planck T+P Bianchi VIIh analysis: anisotropic expansion odds $\sim 1.2 \times 10^5 : 1$ against; $(\sigma_V/H)_0 < 4.7 \times 10^{-11}$ (95%). Constrains *homogeneous* anisotropy; does not identically kill an inhomogeneous spherical CoE, but forbids reading Migkas as global shear.

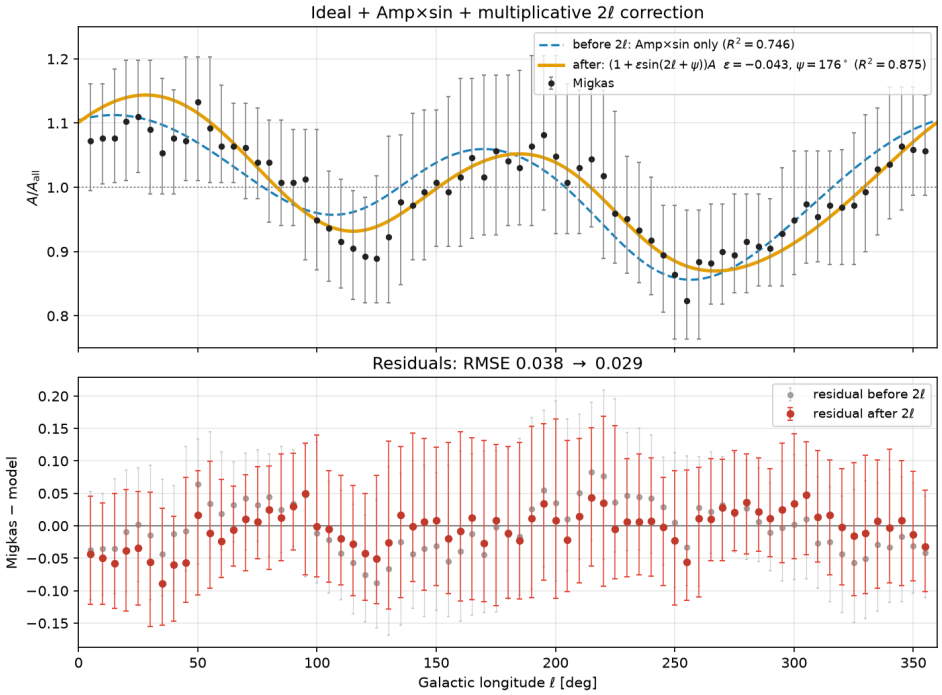


Figure 7: **Key fit figure.** Best ST+Gauss + Amp× sin model with multiplicative 2ℓ correction versus Migkas $A(\ell)$. The geometric backbone is first-principles; Amp× sin and 2ℓ are diagnostic / phenomenological layers. Part of the high R^2 is knobs—stated explicitly.

Migkas et al. (Migkas et al., 2020). Primary empirical target: Fig. 6 A versus Galactic longitude.

8 Discussion: what a good fit does and does not prove

8.1 What $R^2 \approx 0.54$–0.88 means

A geometric ST+Gauss off-center template that correlates with Migkas longitude at $R^2 \approx 0.54$ before knobs is interesting: it is not noise-shaped cherry-picking of a free dipole. Raising R^2 to ~ 0.75–0.88 with Amp× sin and 2ℓ shows that residual phase and quadrupole structure are absorbable—but those layers are knobs. A good fit does *not* prove a real CoE, does not prove that Migkas is cosmological rather than systematic, and does not waive CMB constraints.

8.2 CMB adversary

Planck isotropy (Planck Collaboration, 2020) remains the ceiling. In spherical inhomogeneous models, an off-center observer induces a CMB dipole that forces offsets $\lesssim 15\,$Mpc in standard void constructions (Alnes and Amarzguioui, 2006; Kodama et al., 2010). A CoE *beyond the horizon* is a stronger claim than a Gpc void with us near center: it requires a mechanism by which last scattering looks nearly isotropic while $z \lesssim 0.3$ clusters show $\mathcal{O}(10\%)$ effects. Without that mechanism, the CMB wins. SNIa alone do not tighten the bound (Blomqvist and Mörtsell, 2010).

Saadeh et al. (Saadeh et al., 2016) nearly exclude Bianchi shear. That is **not** the same as excluding an inhomogeneous CoE, but it forbids conflating "anisotropic $H_{\rm obs}$ about us" with

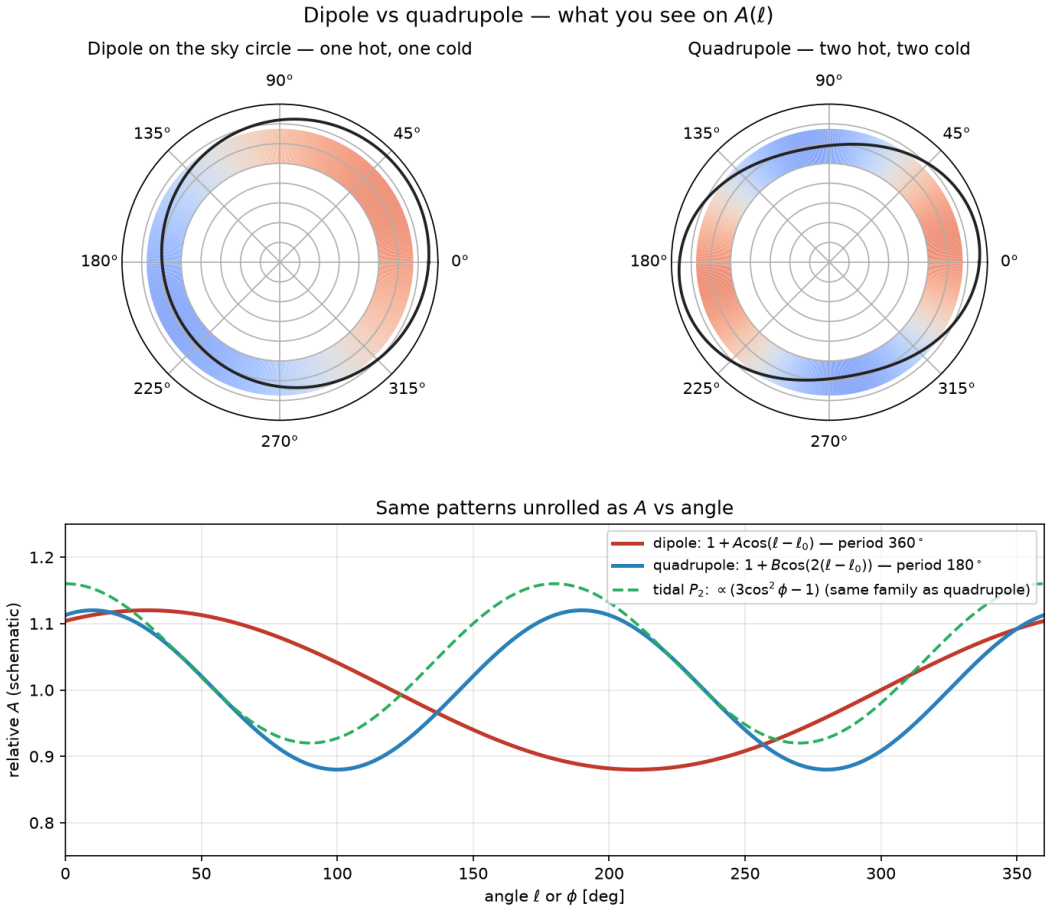


Figure 8: Schematic dipole versus quadrupole patterns on the ST+Gauss sky (bottom: unrolled A versus angle). At small d/D the leading Gauss tidal anisotropy is quadrupole-like (P_2); Migkas longitude strips mix multipoles. Legend coefficients A/B here are schematic multipole amplitudes, not Migkas strip normalizations.

"globally anisotropic expansion."

8.3 Falsifiers

The program is falsifiable, among other ways, by:

1. A demonstrated X-ray / selection systematic that erases Migkas Fig. 6 without a cosmological $A(\ell)$.
2. A CMB-compatible construction that cannot simultaneously host $\mathcal{O}(10\%)$ late-time $A(\ell)$ from a beyond-horizon CoE.
3. Homology regression failure in the forward model ($M = 0$ still anisotropic).
4. Expand/collapse morphology that cannot be reconciled with the observed trough/peak placement once $A_{\rm hratio}$ is fixed.
5. Multipole structure (full-sky C_ℓ) incompatible with cone-scan / longitude-strip data once footprint is modeled honestly.

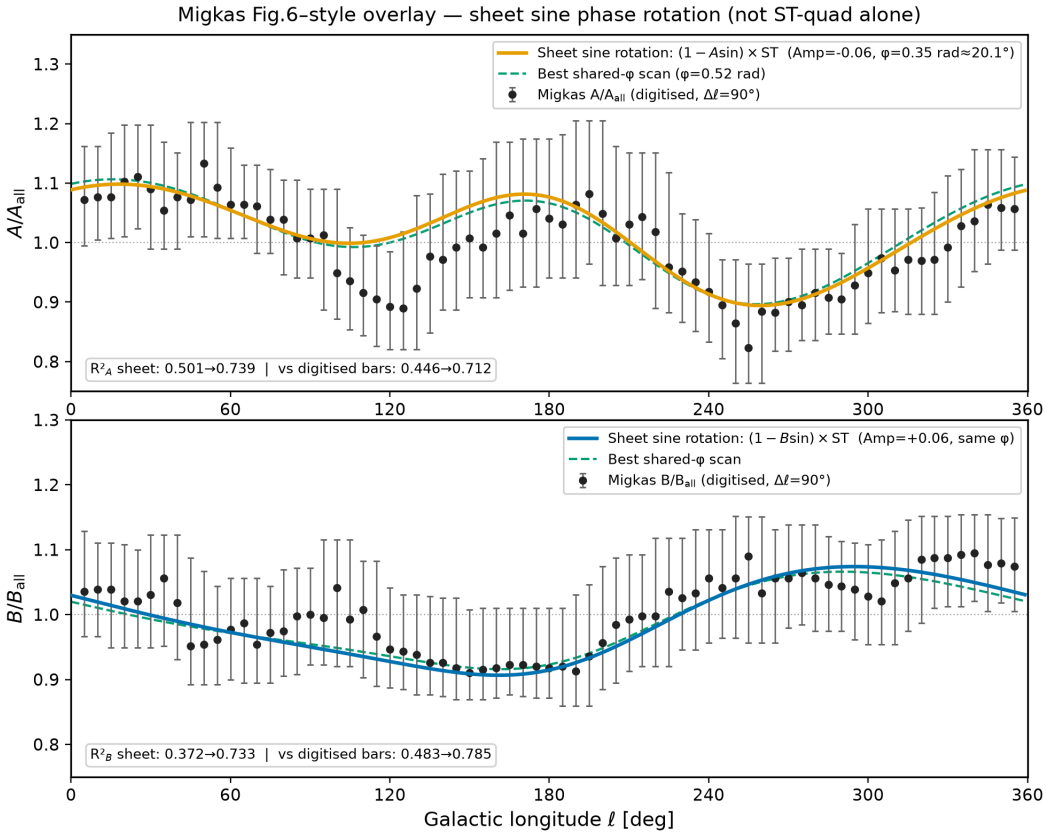


Figure 9: Migkas longitude bars with Amp× sin overlays. **Top:** Migkas A (the quantity used in the R^2 ladder). **Bottom:** Migkas B (ancillary strip normalization; *not* used in Table 1). Phase values on this panel sit on the Amp= −0.06 ridge near $\phi \approx 0.35$ rad; the ladder peak quoted in the text is Amp= −0.06, $\phi \approx 0.20$ rad. Diagnostic methods figure.

8.4 Open problems

No GR metric, no light-travel-time $H_{\rm obs}$, no self-consistent ST+gravity similarity solution, and no beyond-horizon matching that survives Khakshournia-type junction scrutiny (Khakshournia, 2014) are provided here. $\alpha_{\rm eff}$ and κ remain order-unity parameters, not microphysically fixed.

9 Conclusions

We synthesize the Newtonian ST+CoE program as follows:

- Real CoE beyond the horizon; off-center observer in homologous ST expansion or collapse.
- Gauss gravity with decreasing $\rho(r)$ and enclosed mass toward the CoE; $\kappa \sim \mathcal{O}(0.1\text{–}1)$ versus $M_{\rm crit}$ is intended as natural, not extreme.
- Ideal time-integrated gravity is preferred over bolt-on $u \sim \alpha g/H$: integration rescales amplitude only ($\alpha_{\rm eff} \sim 2/3$ with Hubble drag); expand versus collapse flips the $A(\varphi)$ shape.
- Primary Migkas comparison uses $A_{\rm hratio} \propto H_{\rm obs}/H$.
- R^2 ladder: 0.543 (geometry) → 0.746 (Amp× sin) → 0.875 (2ℓ), with Amp= −0.06, $\phi \approx 0.20$ rad, $\varepsilon \approx -0.043$, $\psi \approx 176°$, and top geometry $M/M_0 = 3$, $d/D = 0.5$, $L \approx 0.169$.
- Amp× sin and 2ℓ are diagnostic / phenomenological; nesting dolls are speculative.

- The CMB remains the adversary (Alnes/Kodama $\sim 15\,\mathrm{Mpc}$); Saadeh Bianchi bounds constrain homogeneous shear, not an inhomogeneous CoE identically. This draft does not claim to have falsified the cosmological principle.

Acknowledgments

Conversations with Wolfgang Rindler and Luke Ackerman—especially night conversations after general mechanics and relativity courses—shaped the genesis of this centered-universe program and are gratefully acknowledged. Errors, overclaims, and remaining tensions with the CMB are the author's alone.

References

H. Alnes and M. Amarzguioui, CMB anisotropies seen by an off-center observer in a spherically symmetric inhomogeneous universe, Phys. Rev. D **74**, 103520 (2006), arXiv:astro-ph/0607334.

E. Bertschinger, Self-similar secondary infall and accretion in an Einstein–de Sitter universe, Astrophys. J. **268**, 17 (1983).

M. Blomqvist and E. Mörtsell, Supernovae as seen by off-center observers in a local void, JCAP **05**, 006 (2010), arXiv:0909.4723.

G. Debney, Techniques of global analysis: a cosmological black hole?, NASA TM X-73378 (1977).

A. J. Fennelly, Mon. Not. R. Astron. Soc. **181**, 121 (1977).

A. Heinesen, Multipole decomposition of the general luminosity distance Hubble law, JCAP **05**, 008 (2021), arXiv:2010.06534.

S. Khakshournia, A note on Pathria's model of the universe as a black hole, arXiv:1412.0105 (2014).

H. Kodama, K. Saito, and A. Ishibashi, Analytic formulae for CMB anisotropy for off-center observers in spherical symmetry, arXiv:1004.3089 (2010).

K. Migkas et al., Probing cosmic isotropy with a new X-ray galaxy cluster sample through the L_X–T scaling relation, Astron. Astrophys. **636**, A15 (2020), arXiv:2004.03305.

R. K. Pathria, The Universe as a black hole, Nature **240**, 298 (1972).

Planck Collaboration, Planck 2018 results. VII. Isotropy and statistics of the CMB, Astron. Astrophys. **641**, A7 (2020).

D. Saadeh, S. M. Feeney, A. Pontzen, H. V. Peiris, and J. D. McEwen, How isotropic is the Universe?, Phys. Rev. Lett. **117**, 131302 (2016), arXiv:1605.07178.

J. Smoller and B. Temple, Cosmology with a shock-wave, Commun. Math. Phys. **210**, 275 (1998).

J. Smoller and B. Temple, Shock-wave cosmology inside a black hole, Proc. Natl. Acad. Sci. USA **100**, 11216 (2003), arXiv:astro-ph/0210105.